KAMTEC PHYSICS 53513 SCORING GUIDE 2026

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1

ITEM ONE

- **Aim**

An experiment *to determine the E.m.f of the cell* provided so as *to determine its suitability for use in the radio.*

- **Hypothesis**

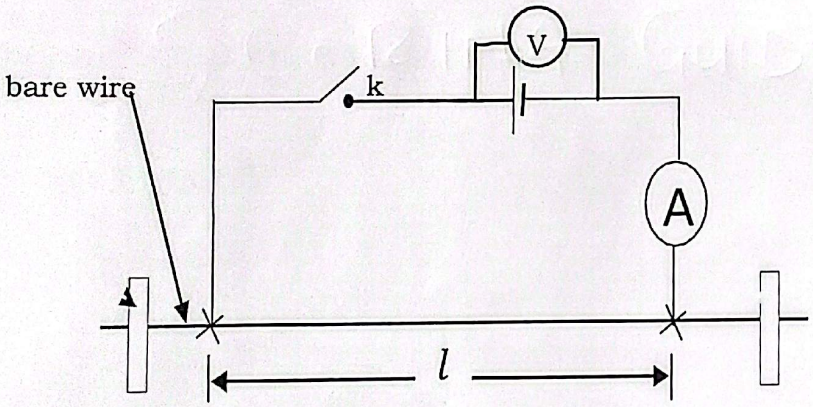
The *E.m.f of the cell* ranges from *1.3V to 1.7V.*

- **Variables**

-Independent variable - Length of the bare wire.
-Dependent variable - Current, I and Voltage, V.
-Controlled/fixed/constant variable - Thickness, temperature of the bare wire.

- **Procedures**

a) The apparatus were arranged as shown below



b) With the switch open, the voltmeter reading, V_0 is taken and recorded.
c) The switch was closed and the length, l, of the bare wire from the zero cm mark of the metre is to point x is measured as 20.0cm.
d) The ammeter reading, I and voltmeter reading, V were taken and recorded.
e) Procedures (c) to (d) were repeated for different values of l = 30.0cm, 40.0cm, 50.0cm,60.0cm and 70.0cm.
f) The results were tabulated in a suitable table
g) A graph of V against I was plotted and the intercept, V_1 on the V axis was determined.
h) The E.m.f of the cell was determined from the expression. $E = \frac{V_1+V_0}{2}$.

- **Possible sources of errors**

- Parallax errors while taking readings from voltmeters and ammeters
- Drained/overused cells
- Loose connections

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➢ **Precautions to the identified errors**

- Positioning the eyes at right angles to the scales of the ammeter and voltmeter being read.
- Using new cells to conduct the experiment.
- Making tight connections.

➢ **Table of Results**

l(cm)	V(V)	I(A)
20.0	0.50	0.40
30.0	0.60	0.36
40.0	0.70	0.32
50.0	0.90	0.28
60.0	1.00	0.24
70.0	1.10	0.20

➢ **Data analysis using graph**

N:B

The graph work is attached behind this booklet.

Intercept, V_1 = 1.65V

E.m.f, $E = \frac{V_1 + V_0}{2}$

$= \frac{1.65+1.50}{2}$

$= \frac{3.15}{2}$

E = 1.58V

➢ **Conclusion and advice**

The E.m.f, E, of the cell is 1.58V, which is in the range of the hypothesis, so I recommend the cell for use in the radio.

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ITEM TWO

Aim

A scientific investigation to determine the mass of the meter rule provided to help confirm whether its suitable for the physics Practical examination

Hypothesis

Mass of the meter rule is between 75g to 140g.

Variables

Independent Variable

- Distance d, of the 100g mass from one end of the meter rule.

Dependent Variables

- Distance y and L, of the center of gravity G of the meter rule and the 100g mass from the knife edge respectively

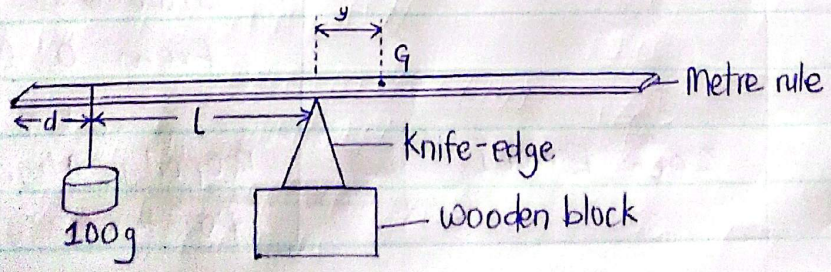
Control Variable

Thickness of the meter rule which may not be Uniform

List of Apparatus:

Piece of thread, pendulum bob, Meter rule, Retord stand with a clamp, stop clock/stopwatch

Experimental set up (Diagram)



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Procedures

a) A block of wood was placed in a suitable table so that it rests on its small cross section area

b) A knife edge was then placed on top of the block

c) The meter rule provided was placed on the knife-edge with its calibrated face upwards

d) The position G at which the metre rule balances was recorded

e) A mass of 100g was suspended at d = 2.0cm mark and the metre rule was adjusted until it balances again

f) The distance y of G from the knife edge and L of the 100g mass were measured and recorded

g) Procedures e) and f) were repeated with the 100g mass hanging at distance d = 6.0, 10.0, 14.0, 18.0 and 22.0cm mark respectively.

h) The results were recorded in a suitable table

I) The slope S of the graph was determined

K) The mass, M of the metre rule was Then Calculated from the expression $M = \frac{100}{S}$

Sources of Errors	Mitigation
- Parallax Errors	Always view the scale of the metre rule from a point directly infront of it.
- Wrong scale	Consistently use the Correct scale that increases from 0 to 100cm
- Zero Errors	Properly Align the zero point.

Table of results

$G = 49.7\ cm$

d (cm)	l (cm)	y (cm)
2.0	26.1	21.6
6.0	23.9	19.8
10.0	21.7	18.0
14.0	19.5	16.2
18.0	17.4	14.3
22.0	15.3	12.4

$$\text{Slope } S = \frac{220 - 113}{266 - 138}$$

$$= \frac{107}{128}$$

$$S = 0.836$$

$$\text{From } M = \frac{100}{S}$$

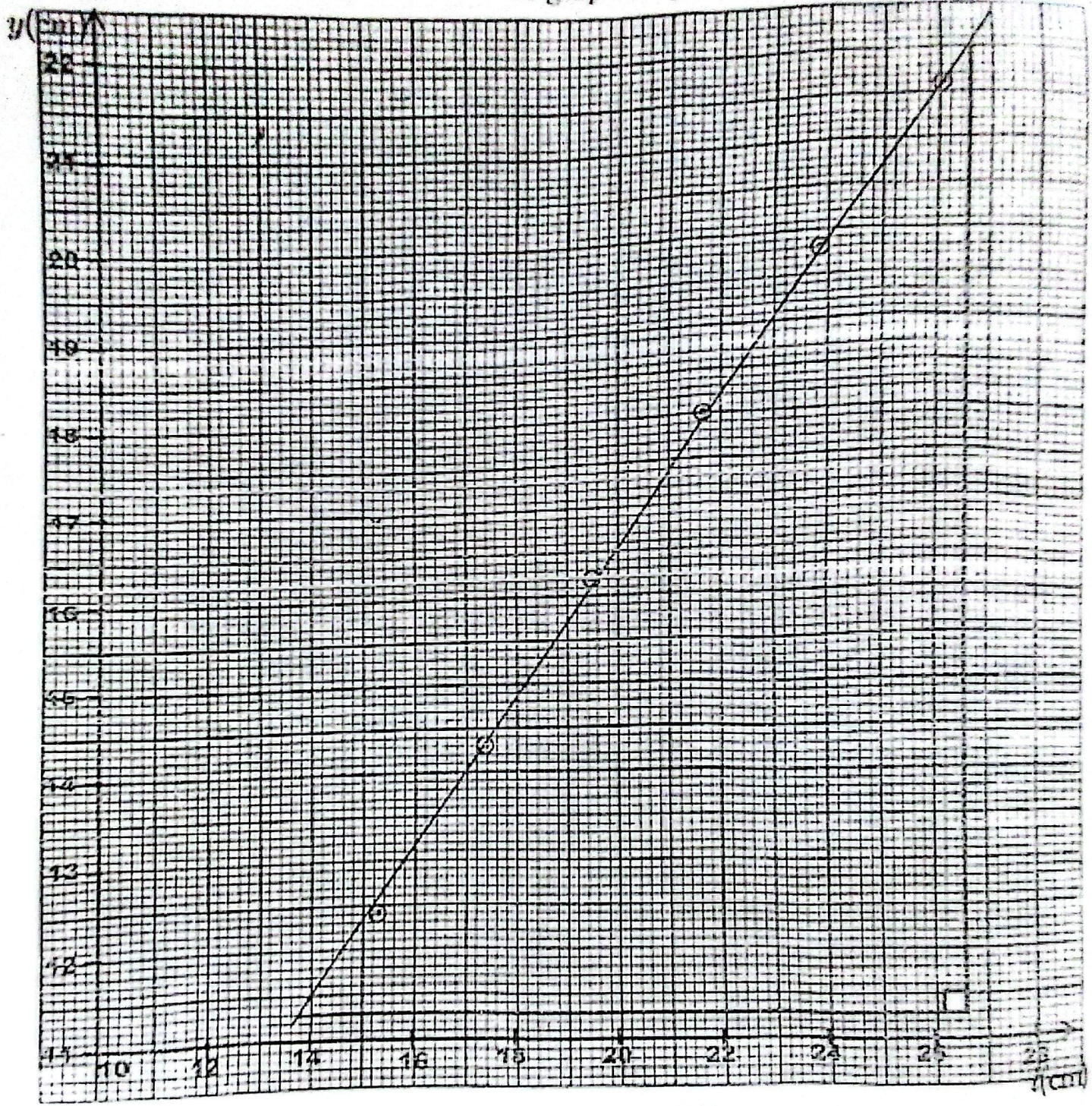
$$= \frac{100}{0.836}$$

$$M = 120g$$

Conclusion and Recommendation

The mass of the metre rule determined to be 120g falls within the provided range Hence, it is suitable for the physics practical examination.

A graph of y against l



535/2,3 PHYSICS P2,3 SCORING GUIDE

S/N	Basis of assessment	Code		Criteria and description	Score	Scoring mode	Total score
1	Aim of experiment	A		Properly stated aim with clear words.	2	A-2	2
				Partially correct aim stated.	1	A-1	
				Incorrect aim or no aim written.	0	A-0	
2	Hypothesis (statement to be proved)	H		Correct hypothesis related to the expt.	1	H-1	1
				Incorrect hypothesis or no hypothesis related to the expt.	0	H-0	
3	Variables	V	Control	Correct control variables stated.	1	V_C-1	3
				Incorrect or no control variables stated	0	V_C-0	
			Independent	Correct independent variable stated.	1	V_I-1	
				Incorrect or no independent variable stated.	0	V_I-1	
			Dependent	Correct dependent variables stated	1	V_D-1	
				Incorrect or no dependent variable stated.	0	V_D-0	
4	Apparatus	A_P		Stating only **all relevant** apparatus	2	A_P-2	2
				Stating **partially relevant** or mixed apparatus.	1	A_P-1	
				Stating only **irrelevant or no** apparatus.	0	A_P-0	
5	Diagram (setup)	D		Complete and well (**fully**) labelled.	2	D-2	2
				Partially labelled	1	D-1	
				No or **wrong diagram** or **no label**	0	D-0	
6	Procedure	P	Relevancy	Relevant/**correct procedure** written	2	P_R-2	4
				Partially relevant or **partially correct** procedure written.	1	P_R-1	
				Irrelevant/**incorrect/no procedure** written	0	P_R-0	
			Coherency	Coherent (logically flowing) procedure	2	P_C-2	
				Partially coherent procedure	1	P_C-1	
				Incoherent procedure	0	P_C-0	
7	Errors or risks identified	E_R		Most of errors (at least 2) identified.	2	E_R-2	2
				A few errors (at least one) identified.	1	E_R-1	
				No error identified.	0	E_R-0	
8	Mitigations (Precautions)	M		Most mitigations (at least 2) stated.	2	M-2	2
				A few mitigations (at least one) stated.	1	M-1	
				No mitigation stated.	0	M-0	

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